

Cool-Nexus: Battery Thermal Management

Fluid selection and cold plate design for an eVTOL battery pack

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Why we did this

- The standard 60/40 EG/water coolant conducts electricity, which is a concern near a high voltage battery pack.
- It also freezes near -45°C , limiting cold weather operation.
- PAO is electrically insulating and remains fluid at low temperatures, so we selected it as the base coolant.
- We asked whether nanoparticles could improve its cooling performance without significantly increasing viscosity.

Picking the fluid

- We found three main PAO nanofluids: Al_2O_3 , hBN, and a MWCNT and MoS_2 hybrid.
- Al_2O_3 and hBN lacked data below 30°C , requiring us to extrapolate.
- The hybrid had measured data across our full range, so we selected it.
- We compiled properties from 20-80 $^{\circ}\text{C}$ and tested the fluids from 20-40 $^{\circ}\text{C}$.

What the properties showed

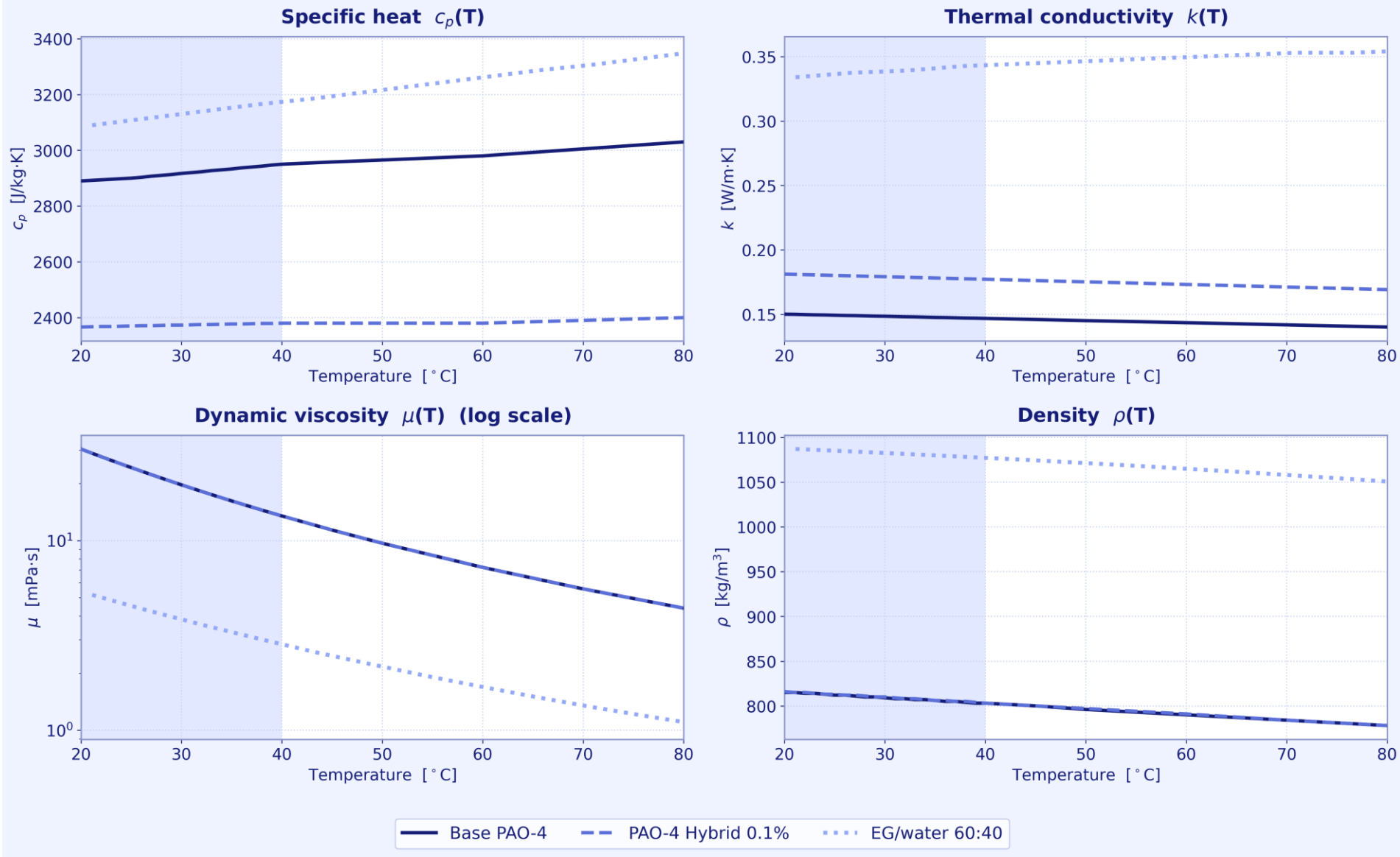


Fig. 1 Specific heat, thermal conductivity, viscosity on a log scale, and density for the three fluids. The shaded band is the 20 to 40 $^{\circ}\text{C}$ test window. In (c) and (d) the base PAO and the hybrid sit on top of one another.

- The hybrid had 21% higher thermal conductivity with nearly unchanged viscosity and density.
- Its specific heat was 18% lower, which was the main tradeoff.
- EG/water performed better thermally but cannot operate at the temperatures we are targeting, so we used it only as a reference.

One number worth flagging

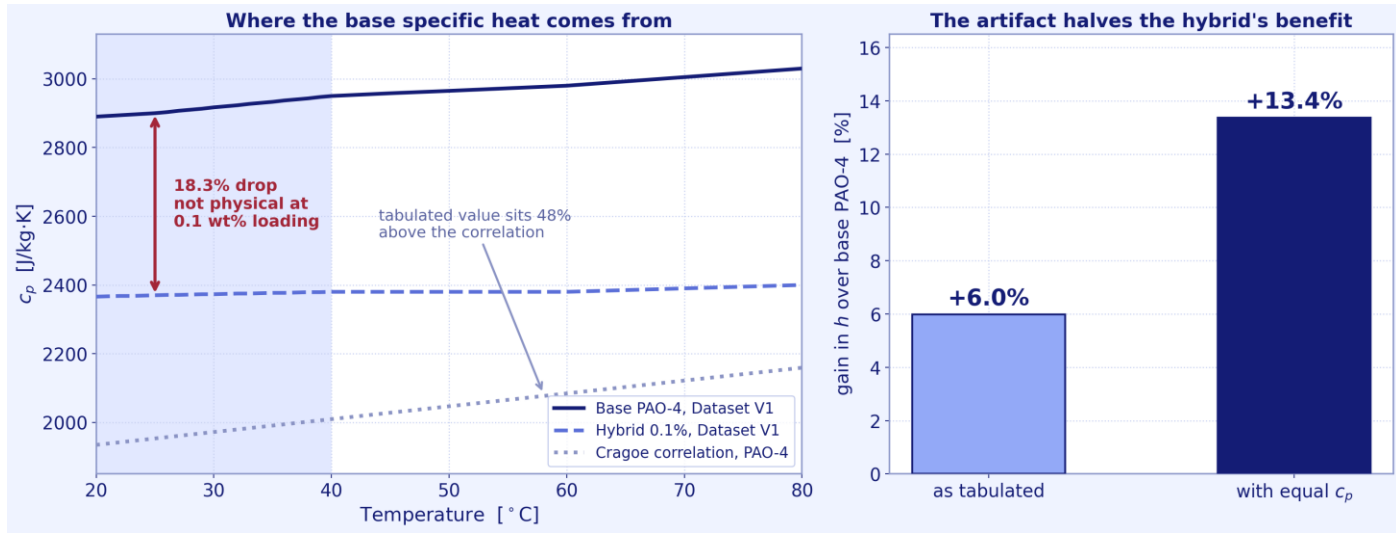


Fig. 2 Left: where the base specific heat comes from. Right: what the hybrid specific heat artifact does to the predicted heat transfer coefficient.

- The 2900 J/kg-K specific heat for base PAO 4 was taken from a published curve.
- The Cragoe correlation gives about 1950 J/kg-K for the same grade.
- We kept the published value but treated the difference as an uncertainty that affects the results.
- The hybrid specific heat also appears questionable. Correcting it increases the predicted benefit from 6% to 13%.

References and acknowledgements

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2. Jaidi, Jeevan, et al. "Performance Study on the Effect of Coolant Inlet Conditions for a 20 Ah LiFePO4 Prismatic Battery with Commercial Mini Channel Cold Plates." Electrochem, vol. 3, no. 2, MDPI, May 2022, pp. 259–75.
3. Pournorouz, Zahra. "In Situ Nanostructured Nanofluids for Thermal Management in Hypersonic and Electrified Aircraft." 2026. Unpublished research proposal, Stevens Institute of Technology.
4. Wu, Zhiwei, et al. "Research on Battery Heat Generation Characteristics and Thermal Management System Applied to a Typical EVTOL." Applied Thermal Engineering, vol. 257, Pergamon, Aug. 2024, p. 124187.

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Designing the cold plate

- PAO is viscous, so limiting pressure drop was a major design goal.
- We compared three layouts using the same fluid properties, flow rate, and heat load.
- Design 3, a U plus U manifold with six 30×6 mm channels, performed best.
- We modeled it in SolidWorks on a 250×500 mm plate and analyzed it in Ansys Discovery.

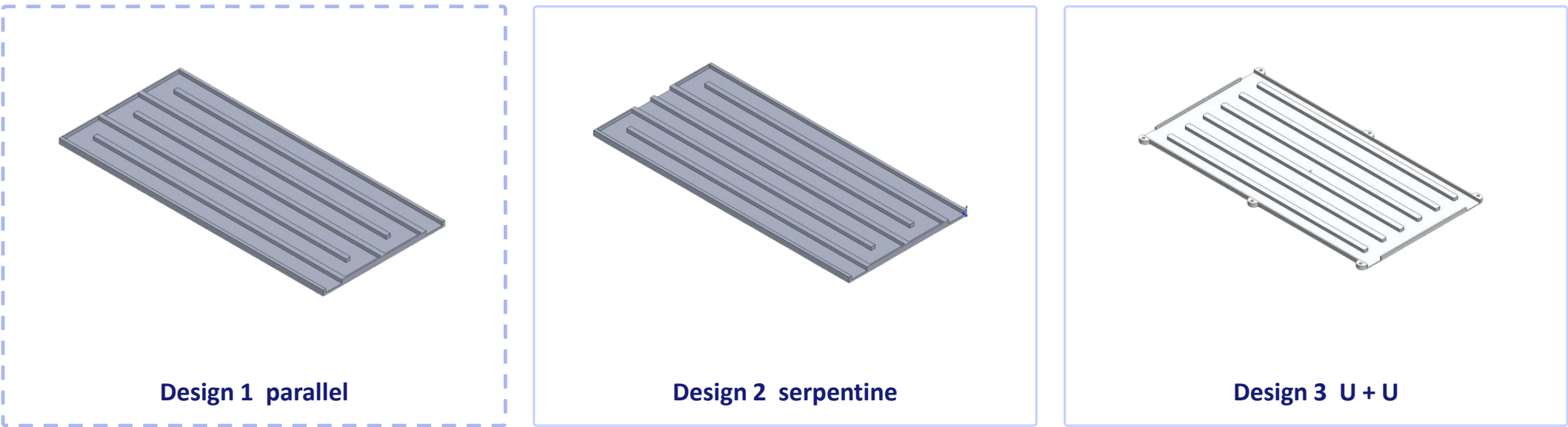


Fig. 3 The three channel layouts we compared by hand before committing to one.

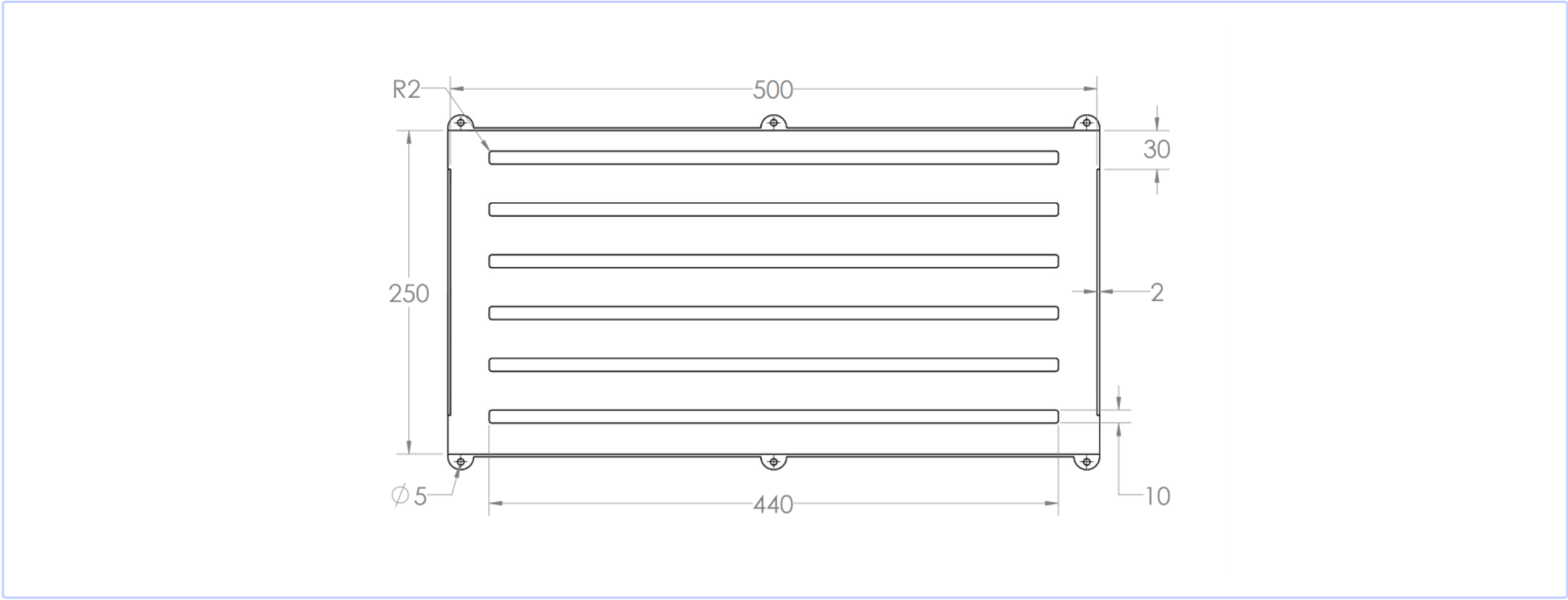


Fig. 4 Design 3 as built. Six parallel 30×6 mm channels in a 500×250 mm aluminium 6061 plate, fed by a U+U manifold. The flow is deeply laminar at $Re \approx 26$.

Setting	Value
Heat load	500 W (18 cells, landing hover peak)
Inlet temperature	20 $^{\circ}\text{C}$
Total mass flow	0.068 kg/s baseline, 0.05 and 0.10 sweep
Channel length	0.47 m, straight section only
External walls	near adiabatic, $h = 1 \text{ W/m}^2\cdot\text{K}$
Plate material	Aluminium 6061
Requirements	T max $\leq 40^{\circ}\text{C}$, $\Delta T \leq 5^{\circ}\text{C}$

Does the model agree with the CFD?

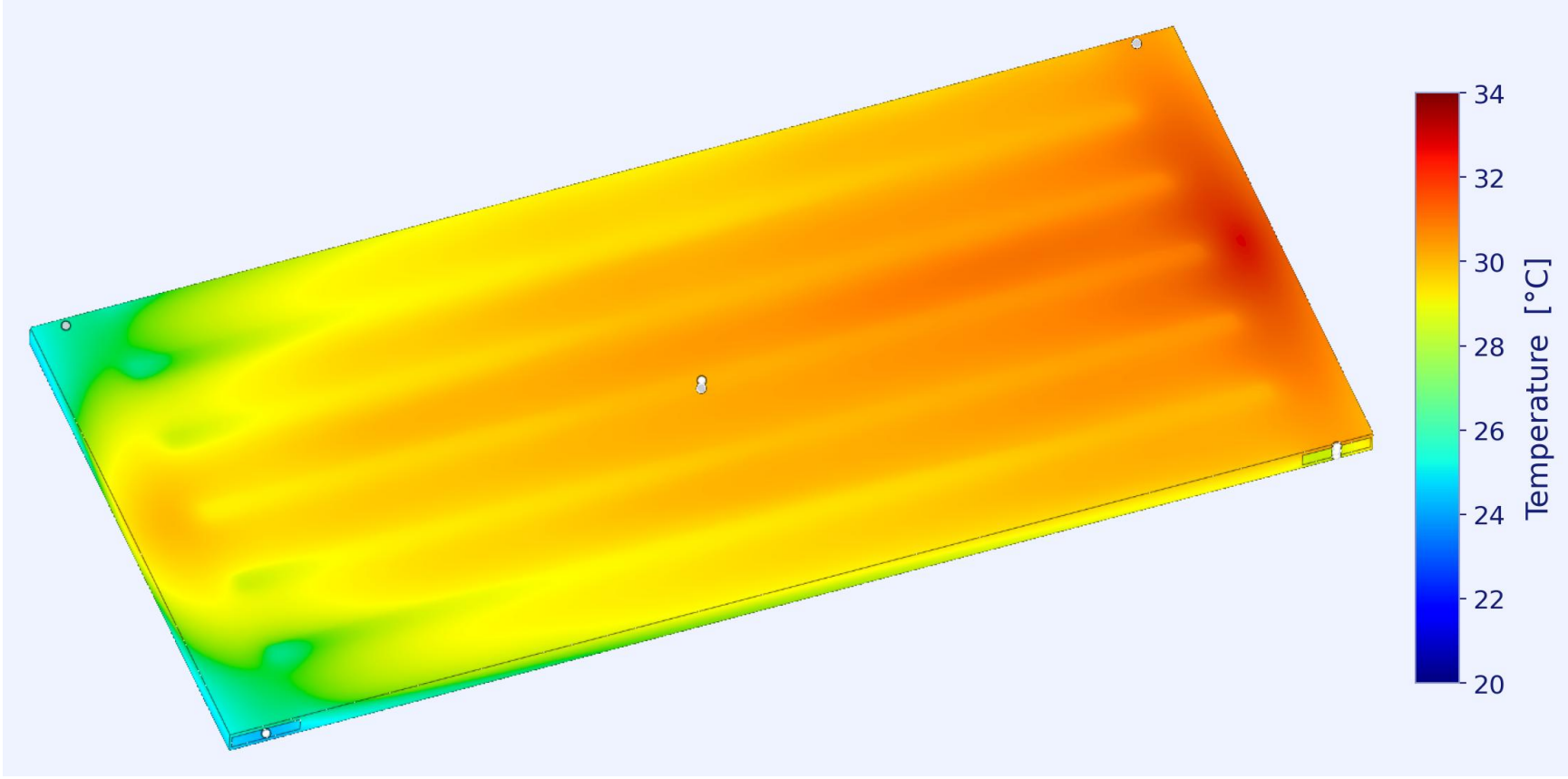


Fig. 5 Surface temperature at 0.068 kg/s and 500 W. The hottest point is the outlet side corner. The cool inlet plenum sits outside where the cells are, which is why uniformity is measured over the cell footprint.

Quantity	MATLAB	CFD	Diff
Outlet temperature	22.54 $^{\circ}\text{C}$	22.5 $^{\circ}\text{C}$	0.2%
Heat into the fluid	500 W	500 W	0%
Channel wall temp (avg)	33.86 $^{\circ}\text{C}$	27.0 $^{\circ}\text{C}$	22.54%
System pressure drop	399 Pa	343 Pa	15.1%
Max plate temperature	33.86 $^{\circ}\text{C}$	34.0 $^{\circ}\text{C}$	0.41%

- The hand calculation and CFD report different wall temperatures because they measure different things.
- The hand calculation gives an average channel wall temperature, while Discovery gives the highest local plate temperature.
- We therefore compared outlet temperature and pressure drop for validation.
- Maximum temperature was evaluated separately as a design limit.

Results

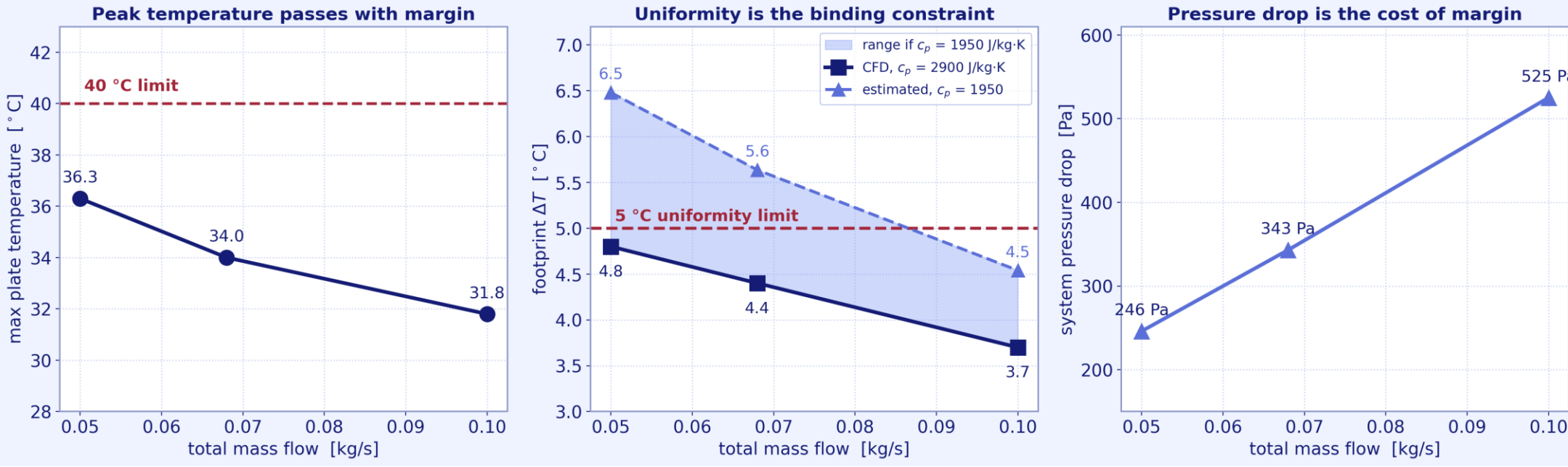


Fig. 6 Flow sweep against both requirements. Uniformity is plotted on its own axis. The shaded band shows the effect of the specific heat uncertainty.

- The design met both requirements at every flow rate tested.
- At 0.05 kg/s, temperature uniformity passed by only 0.2 $^{\circ}\text{C}$, where flow maldistribution was also greatest.

The assumption that did not hold

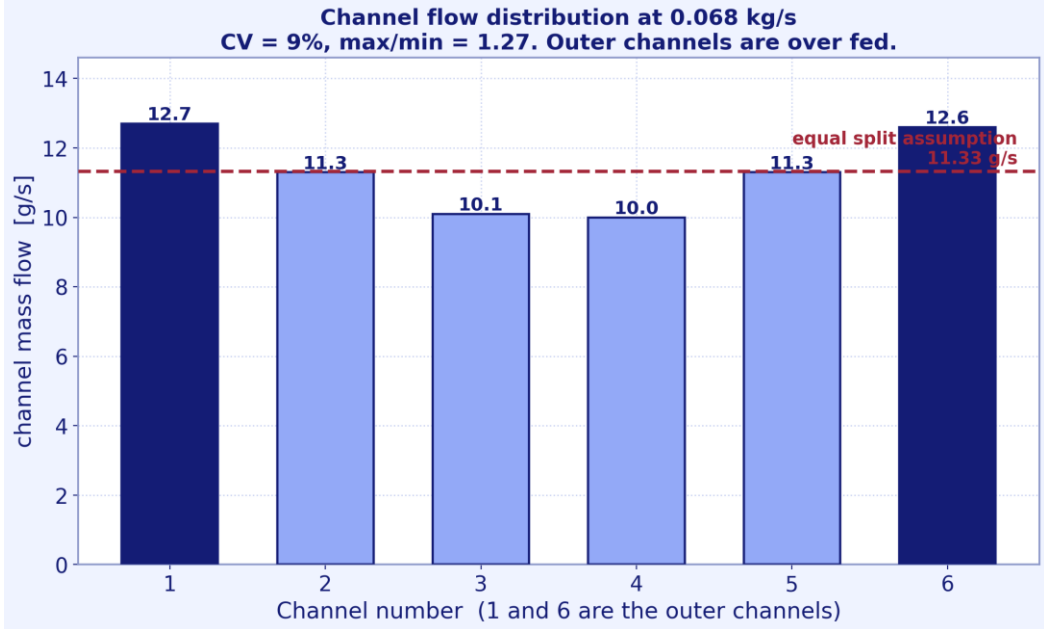


Fig. 7 CFD predicted flow through each channel at 0.068 kg/s. No flow measurements were taken.

- Our hand calculations assumed equal flow through all six channels.
- CFD showed that the outer channels carry about 27% more flow than the inner channels.
- The coefficient of variation was 9% at baseline flow and 13% at low flow.
- The uneven flow is caused by the pressure gradient in the manifold, not by differences in channel length.

Conclusion

Requirement	Target	Result	
Heat load	500 W	500 W	PASS
Interface temperature	$\leq 40^{\circ}\text{C}$	34.0 $^{\circ}\text{C}$	PASS
Uniformity ΔT	$\leq 5^{\circ}\text{C}$	4.4 $^{\circ}\text{C}$	PASS

- Temperature uniformity is the main constraint, with 6 $^{\circ}\text{C}$ of temperature margin but only 0.6 $^{\circ}\text{C}$ of uniformity margin.
- The result depends strongly on the uncertain PAO specific heat.
- Using 1950 J/kg-K increases the fluid temperature rise by about 1.2 K, causing the baseline uniformity to increase from 4.4 $^{\circ}\text{C}$ to about 5.6 $^{\circ}\text{C}$ and fail the 5 $^{\circ}\text{C}$ limit.
- Only the 0.10 kg/s case would still pass.

Future Work

- Apply CDF analysis to Application Definition (Plate Size: 294mm x 242mm)
- Test plate with the battery enclosure
- Find more data on other nanofluids
- Explore different geometries
- Use Ansys Fluent for CFD
- Analyze in more Extreme conditions

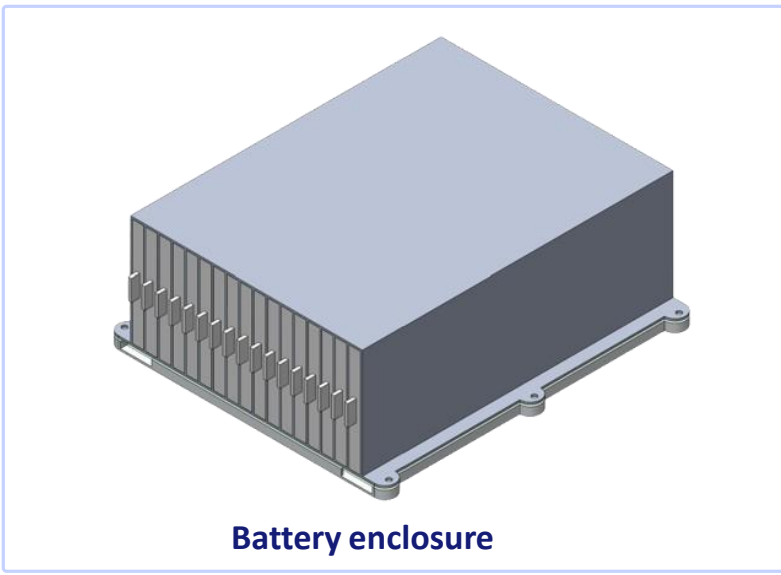


Fig. 8 The enclosure sized to the 493 W and the printed model built for manufacturability.

My Contributions

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